Matthew Schwartzman

Address: Department of Economics
Yale University
New Haven, CT 06520-8268

Telephone: +1 (773) 256-8626

E-mail: matthew.schwartzman@yale.edu

Web page: www.matthewschwartzman.com

Citizenship: Canadian (US F-1 visa)

Fields of Concentration:
Primary Fields: Macroeconomics, Economic Development
Secondary Field: Labor Economics

Comprehensive Examinations Completed:
2021 (Oral): Macroeconomics, Economic Development
2020 (Written): Microeconomics, Macroeconomics

Dissertation Title: *Worker and Consumer Choice in the Process of Structural Transformation*

Committee:
Professor Michael Peters (Co-Chair)
Professor Fabrizio Zilibotti (Co-Chair)
Professor Kevin Donovan

Education:
Ph.D., Economics, Yale University, 2025
M.Phil., Economics, Yale University, 2022
M.A., Economics, Yale University, 2020
B.A., Economics and Mathematics, University of Toronto, 2018

Fellowships, Honors and Awards:
Sylff Fellowship, Yale University, 2021-24
Graduate Fellowship, Yale University, 2019-present
Cowles Foundation and Economic Growth Center Fellowship, Yale University, 2019-24
University College Medal for Best Degree in the Arts, University of Toronto, 2018

Research Grants:
Structural Transformation and Economic Growth (STEG) PhD Research Grant, "Do Workers, Firms, or Consumers Drive the Structural Transformation of Retail?" (£7,835), 2023-24

Teaching/Mentorship Experience:

Fall 2023: Teaching Assistant to Prof. William Nordhaus, Intermediate Macroeconomics (undergraduate), Yale College

Summer 2022 & Summer 2023: Co-Director, Herb Scarf Summer Research Opportunities, Yale College

Fall 2022 – Spring 2023: Teaching Assistant to Profs. Rebecca Toseland and Giovanni Maggi, The Senior Essay (undergraduate), Yale College

Spring 2022: Teaching Assistant to Prof. Michael Peters, Intermediate Macroeconomics (undergraduate), Yale College

Fall 2021: Teaching Assistant to Prof. Fabrizio Zilibotti, Intermediate Macroeconomics (undergraduate), Yale College

Summer 2021: Teaching Assistant to Prof. Vitor Possebom, Introductory Econometrics (undergraduate), Yale College

Research Experience:

Graduate Research Assistant to Prof. Michael Peters, Yale University, 2020-21
Pre-Doctoral Research Assistant to Prof. Alessandra Voena, University of Chicago, 2018-19
Undergraduate Research Assistant to Prof. Michael Baker, University of Toronto, 2017-18
Undergraduate Research Assistant to Prof. Arthur Blouin, University of Toronto, 2016-18

Working Papers:

"From Street Markets to Shopping Malls: Technology and Wedges in the Transformation of the Service Sector" (2024), *Job Market Paper*

"Risk Management and Aggregate Migration Flows" (2023)

Work In Progress:

"A Theory of Endogenous Degrowth and Environmental Sustainability", with Philippe Aghion, Timo Boppart, Michael Peters, and Fabrizio Zilibotti

"Higher Education as an Engine of Service-Led Growth," with Alvaro Cox

Seminar and Conference Presentations:

STEG Theme 2 Workshop, NYU Abu Dhabi, January 2024, "From Street Markets to Shopping Malls: Technology and Wedges in the Transformation of the Service Sector"

RIDGE Forum Towards Sustainable Growth, Universidad de los Andes, December 2022, "Risk Management and Aggregate Migration Flows"

Referee Service:

Review of Economic Dynamics

Languages:

English (native), French (fluent)

References:

Prof. Michael Peters
Yale University
Department of Economics
New Haven, CT 06520
87 Trumbull Street, Office B221
Phone: 203-436-8475
m.peters@yale.edu

Prof. Fabrizio Zilibotti
Yale University
Department of Economics
New Haven, CT 06520
87 Trumbull Street, Office B230
Phone: 203-432-9561
fabrizio.zilibotti@yale.edu

Prof. Kevin Donovan
Yale University
School of Management
New Haven, CT 06511
2512 Evans Hall, 165 Whitney Ave
kevin.donovan@yale.edu

Dissertation Abstract

Economic development encompasses more than just growth of average productivity and income per capita. Growth cannot occur without a broader process of structural transformation that reallocates economic activity across production units, sectors, and locations. This reallocation depends on workers' choices of where and how to supply their labor and consumers' choices of what to consume. The three chapters of my dissertation examine different forces that influence worker and consumer choices – such as labor market frictions, market access costs, uninsurable risk, and technological spillovers – and these choices' implications for the aggregate path of the economy.

"From Street Markets to Shopping Malls: The Transformation of the Service Sector" [Job Market Paper]

Services are the largest employment sector in the world, and recent studies show that it is a significant source of productivity growth in developing countries. But little is understood about how the service sector can generate this growth, especially in low-income settings, where it primarily consists of basic consumer services like retail and food service. I argue that the key driver of service-led growth in developing countries is the service transformation: the ongoing process of modern service firms like supermarkets and restaurants replacing traditional micro-entrepreneurs like street vendors and food hawkers.

I use micro-data to show that this process reflects two underlying forces: workers moving out of subsistence employment in traditional services to more productive modern jobs, and consumers increasing their relative demand for modern services as they get richer. I build these two forces into a general equilibrium model of the service sector. The model micro-founds income effects on consumer demand with fixed shopping costs to access modern services. It also features search frictions for wage work in the labor market, which micro-found the marginal productivity gap between modern and traditional services. Together, these features imply that small improvements in the technology used by modern service firms set off a cycle of amplified productivity growth. As workers transition to the modern sector, they increase aggregate income and redirect demand to modern services, which pulls in more workers and generates further growth. A significant portion of this growth comes from reallocation of workers to more productive jobs, rather than mechanical growth from making workers more productive in the jobs they already hold.

I estimate the model with Brazilian micro-data on consumer service expenditures, worker earning and employment dynamics, and the local allocation of service employment and income. I validate the estimated model by testing it against a number of untargeted moments, including a natural experiment from Brazil's trade liberalization. Counterfactual simulations of the estimated model show that the main catalyst of Brazilian service transformation from 2000 to 2010 was improving technology in the modern service sector. Demand effects amplified this transformation by a factor of 1.22. The resulting transformation generated service-led productivity growth. Over half of this growth was reallocation gains from shifting service workers into the more productive modern sector.

"A Theory of Endogenous Degrowth and Environmental Sustainability", with Philippe Aghion, Timo Boppart, Michael Peters, and Fabrizio Zilibotti

Climate activists often argue that economic growth must slow down or reverse course to prevent an environmental catastrophe. We assess this argument in light of ongoing structural transformation, recognizing that growth is not simply a scale-up of production, but a change in what is produced. We build a model with non-homothetic preferences and innovation that can be directed towards reducing the cost of material production or improving the quality of final goods. As real income grows, consumers demand higher-quality goods, and innovation shifts towards increasing the quality rather than the quantity of consumer goods. We associate higher quality with a higher service input, so that over time the economy becomes more reliant on service labor and uses environmentally harmful material inputs less intensively. If quality improvements are incompletely measured, the shift to quality-led growth will appear in data as a growth slowdown. We use aggregate and micro data to discipline the model and find that it fits cross-sectional and time-series patterns of the US economy. The calibrated model shows that innovation subsidies can accelerate the transition to quality-led growth and prevent excessive environmental damages.

"Risk Management and Aggregate Migration Flows"

I study whether the need to manage uninsured risk influences the destination choice of migrants in low-income settings. I develop a parsimonious model of migration as a risk management strategy. Risk-averse households have an incentive to migrate between origin-destination pairs whose economic shocks have low covariance, because these location pairs are better hedges for one another. Under common preference specifications, the model generates a risk-augmented gravity equation that I estimate using migration data from the Philippines. Empirical results confirm that migration is substantially higher between origin-destination pairs with low covariance of shocks.